

Project proposal

Smart Helmet with IoT

A sensing and emergency-alert helmet that makes motorcycle riding safer.

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1. Summary

This project proposes a smart motorcycle helmet built around an ESP32 controller. It checks the rider before the ride, watches for danger during the ride, and after a crash it judges how severe the impact was and sends an alert with the rider's location and photo evidence.

2. Problem

- Riders may start without the helmet properly fastened, or after drinking.
- Drowsiness reduces reaction time on long rides.
- Vehicles approaching from behind can sit in the rider's blind spot.
- Wet roads and potholes are hard to judge in time.
- After a crash, help can be delayed because nobody knows it happened, where, or how serious it is.

3. Objectives

- Detect crashes automatically and send an emergency SMS with location and a severity level.
- Reduce risky starts through helmet-worn and breath-alcohol checks.
- Warn the rider of drowsiness and of vehicles in the blind spot.
- Keep a crash record (sensor log and camera frames) that survives the impact.
- Collect road-condition data, such as potholes and wet roads, for later analysis.

4. Proposed solution

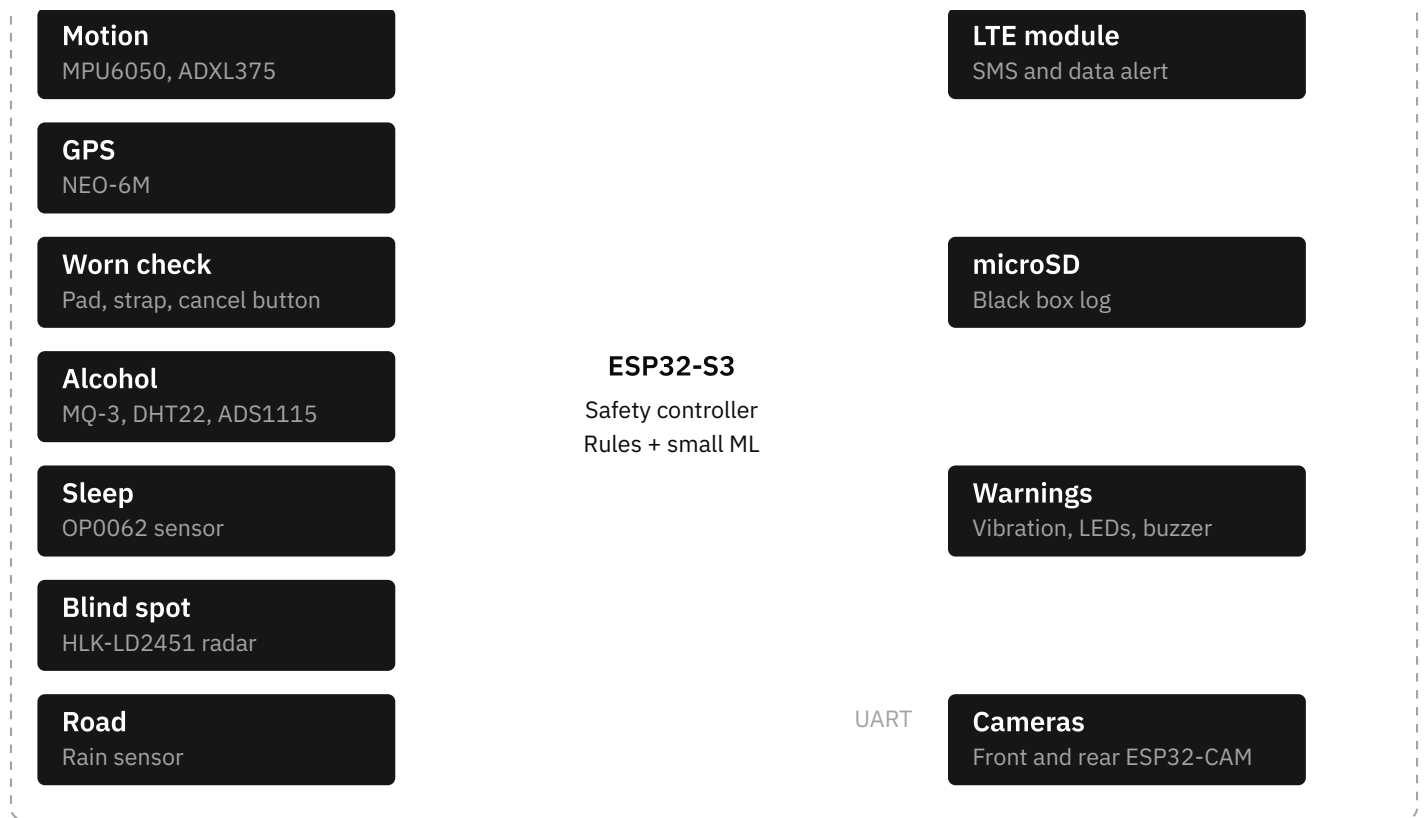
The core build has eight features. Two cameras and road-condition analysis are added on top.

Risk	Feature	Key components	Approach
Helmet not worn properly	Worn check	Capacitive pad, chin-strap switch	Two signals must agree before the helmet reports "worn".
Drinking and riding	Alcohol check	MQ-3 gas sensor, DHT22	Breath screening before the ride, shown as advice on the helmet.
Unreported crash	Crash detection	ADXL375, MPU6050, GPS	Impact, tilt and speed checks, then a countdown the rider can cancel.
Severity unknown	Severity level	ADXL375, MPU6050, GPS	Minor, moderate or severe from impact, speed, orientation and stillness.
Help not called	Emergency alert	4G LTE module	SMS with location and severity, then data and images.
No crash record	Black box and cameras	microSD, front and rear ESP32-CAM	Continuous log plus camera frames saved on a crash.
Drowsy riding	Sleep detection	OP0062 sensor	Measures eye closure and warns by vibration and tone.
Vehicles behind	Blind spot assist	HLK-LD2451 radar, MPU6050	Detects approaching vehicles and warns by vibration and LEDs.
Bad roads	Road-condition logging	Front camera, GPS, rain sensor	Logs potholes and wet-road risk with location for later analysis.

5. System architecture

Cloud and phone

SMS, data, map



The ESP32-S3 samples every sensor in separate tasks, makes safety decisions with fixed rules, and drives the outputs. The two ESP32-CAM boards are separate so a camera power dip cannot reset the crash logic. The LTE module sends alerts to the cloud and to the rider's emergency contact.

Crash response

1. Speed, helmet-worn and impact checks trigger.
2. Tilt and stillness confirm the crash.
3. A buzzer and vibration countdown starts, and the rider can cancel.
4. If not cancelled, severity is scored and the SMS is sent.
5. Camera frames and black-box data are saved and uploaded, with retries until delivery succeeds.

Safety-critical decisions use fixed rules so they behave predictably. Machine learning is limited to uncertain patterns, such as drowsiness and wet roads, and it never acts alone on a crash alert. Heavier analysis, such as confirming potholes in camera frames, runs offline on the logged data.

Severity level. The helmet scores each crash from the impact sensor using a simplified impact score (peak acceleration and how long it lasts), together with speed before impact, final orientation, stillness and the rider's response. It is not a medical head injury criterion, because a sensor on the helmet shell is not a standard headform measurement. The level is a triage hint for responders, not a

medical assessment.

6. Components and budget

Category	Items	Cost (Rs)
Controller and power	ESP32 board, Li-ion cell, charger, 5 V boost converter	2,060
Sensing	MPU6050, GPS, helmet-worn pad, MQ-3, DHT22, ADS1115, radar	5,011
Communication and storage	4G LTE module, microSD module	4,580
Cameras	Two ESP32-CAM modules	4,300
Total of priced parts		15,951

Prices are Sri Lankan retail listings seen in October 2026 and may change. Still to be costed: the ADXL375 impact sensor, the OP0062 sleep sensor, a SIM plan, a microSD card, vibration motors, LEDs and buzzer, switches, wiring and an enclosure. The budget uses a standard ESP32 board, and the ESP32-S3 is to be confirmed.

7. Evaluation

Each feature will be tested and the results reported as numbers, including false alarms and missed detections.

- **Crash detection and severity:** detection rate, false alarms and correct level in drop, pendulum and simulated-fall tests.
- **Emergency alert:** time from crash to SMS delivery.
- **Alcohol check:** warm-up time and repeatability.
- **Sleep detection:** detection of eye closures across several riders, with and without glasses.
- **Blind spot:** hits and misses in rides with a following vehicle, with the head straight and turned.
- **Power:** battery runtime in each operating state.

8. Limitations and responsible use

- This is a prototype and not certified protective equipment

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- The alcohol check is a screening aid and never affects the engine.
- Early testing uses drop rigs and dummies, not real crashes.
- Consent is required before recording or sharing location, images or audio.